Richard B. Anderson

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EDUCATION

The University of Texas at El Paso, El Paso, TX *May 2026*
Bachelor of Science in Physics, Magna Cum Laude
Bachelor of Science in Mathematics, Magna Cum Laude

- GPA: 3.81 / 4.00
- **Outstanding Senior in Physics:** UTEP Department of Physics, 2026. Departmental honor recognizing outstanding undergraduate achievement in physics.

RESEARCH EXPERIENCE

Undergraduate Research Assistant *El Paso, TX*
UTEP Astrophysics Theory Group | Advisor: Dr. Kedron Silsbee *Aug 2022 – May 2026*

- Investigated whether planetesimals large enough to subsequently grow through pebble accretion can be captured by the time-varying gravitational potential of collapsing prestellar cores.
- Jointly developed analytic estimates of capture volumes with Dr. Kedron Silsbee and tested them against numerical simulations for three one-dimensional collapse models.
- Redesigned and ported the simulations from Python to C, implementing inter-node and intra-node parallelization using MPI and OpenMP on UTEP's high-performance computing cluster.
- Optimized numerical integration, interpolation, and root-finding algorithms and exploited physical constraints to reduce simulation runtimes from about eight hours to five minutes, a nearly 100-fold speedup.
- Averaged over more than 60 million planetesimal trajectories for a single collapse model and combined capture volumes with velocity and size distributions to assess captured bodies as seeds for planet formation.

Undergraduate Research Intern *Arlington, TX*
UT Arlington | Neutrino and Rare Event Searches *May 2024 – Dec 2024*

- Simulated alpha-particle interactions in Geant4/NEXUS and analyzed NEXT-100 commissioning data to study how electron-ion recombination, electron diffusion, and electron lifetime shape the detected signals.
- Developed Python/SciPy waveform analyses and presented preliminary results to the NEXT collaboration.
- Calculated electron energy and angular distributions predicted by different effective field theory contributions to neutrinoless double-beta decay using nuDoBE.
- Applied Kolmogorov-Smirnov tests to identify which operator signatures the experiment could distinguish and which are degenerate, assessing its ability to discriminate underlying decay mechanisms.

SULI Intern, Nonproliferation and National Security Department *Upton, NY*
Brookhaven National Laboratory *Jun 2025 – Aug 2025*

- Developed a coupled simulation framework combining LIGGGHTS discrete element modeling of fuel pebble motion with Monte Carlo neutron transport in OpenMC to generate burnup-dependent spectra for nondestructive reactor monitoring.
- Identified and implemented a virtual-lattice method that reduced geometry memory requirements, allowing more realistic pebble-bed reactor configurations to be modeled.
- Developed machine-learning pipelines for predicting fuel burnup from simulated gamma-ray spectra and explored transformer-based architectures for extracting information from the full spectral distributions.

Undergraduate Research Assistant *El Paso, TX*
UTEP Aerospace Center | Missile Systems Innovation Team *Sep 2023 – May 2024*
- Built a high-fidelity missile-sizing simulation to evaluate performance across subsonic, transonic, and supersonic flight regimes for specified payload and range configurations.
- Trained computer-vision models for the missile seeker and integrated C++/OpenCV software with ArduPilot and UAV hardware for onboard testing.
- Worked across computer-vision and model-based systems engineering teams to integrate computational tools and simulation infrastructure.

PUBLICATIONS & PRESENTATIONS

Manuscript: Richard B. Anderson and Kedron P. Silsbee. *Gravitational Capture of Large Interstellar Planetesimals by Collapsing Prestellar Cores.* Submitted for publication. arXiv:2611.00537.
Conference Presentation: Richard B. Anderson and Kedron P. Silsbee. *Capture of freely-floating planetesimal into protoplanetary disks.* Fall 2025 Joint Meeting of the Texas, Arkansas, Oklahoma Section of the APS, Texas Section of the AAPT & Zone 13 of the SPS, October 10, 2025. Abstract FB01.00063.

TEACHING, LEADERSHIP & OUTREACH

Undergraduate Teaching Assistant, Electromagnetism Laboratory *El Paso, TX*
The University of Texas at El Paso, Department of Physics *Aug 2025 – Dec 2025*
- Guided students through laboratory setup, data collection, and analysis while connecting measurements to electromagnetism concepts.
- Diagnosed experimental and conceptual problems and coached students in uncertainty analysis and technical communication.

Academic Tutor *El Paso, TX*
El Paso Community College, Student Success *Jul 2022 – Aug 2023*
- Tutored mathematics from remedial algebra through differential equations in one-on-one and online settings.
- Adapted explanations to varied preparation levels, emphasizing problem formulation, step-by-step reasoning, and independent problem solving.

Scientific-Computing Module Developer and Instructor *El Paso, TX*
The University of Texas at El Paso
- Designed a two-session module introducing approximately a dozen undergraduates to the use of artificial intelligence tools in scientific computing.
- Guided students in using 131 radial-velocity observations of Gamma Cephei to construct a Keplerian model, evaluate its fit with chi-squared, and refine it with a three-body simulation.
- The modules are the subject of a manuscript in preparation by UTEP's College of Education.

Co-founder and President *El Paso, TX*
UTEP Astronomy Club
- Co-founded the club to connect students from diverse backgrounds with astronomy through hands-on observing and collaborative learning.
- Organized public observing nights and presentations that introduced El Paso community members, including first-time participants, to astronomy.

HONORS & AWARDS

NSF Graduate Research Fellowship Program Honorable Mention: National Science Foundation *2026*
National recognition of research potential and broader impacts in the GRFP competition.

Terry Foundation Scholarship: Terry Foundation *2023–2027*
Full-ride scholarship recognizing academic achievement, leadership, and service.

NuPEERS Nuclear Physics Summer School: NuPEERS *2023*
Undergraduate summer training in nuclear physics.

TECHNICAL SKILLS

Programming: Python, C, C++, Fortran, Mathematica, Bash
Scientific Computing: NumPy, SciPy, Pandas, HDF5, TensorFlow
Simulation: REBOUND, Geant4/NEXUS, OpenMC, LIGGGHTS
HPC & Tools: Linux, SLURM, MPI, OpenMP, CUDA, Git, CMake, Docker, ParaView, MATLAB